

Home Test Solution for ModulrFinance Opportunity

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Purpose

Document is created as a solution for 3 home assignments. It includes detailed analysis performed on different API's available in the Modulr. Embedded with project feature file, step Definition, step implementation and API excel sheet.

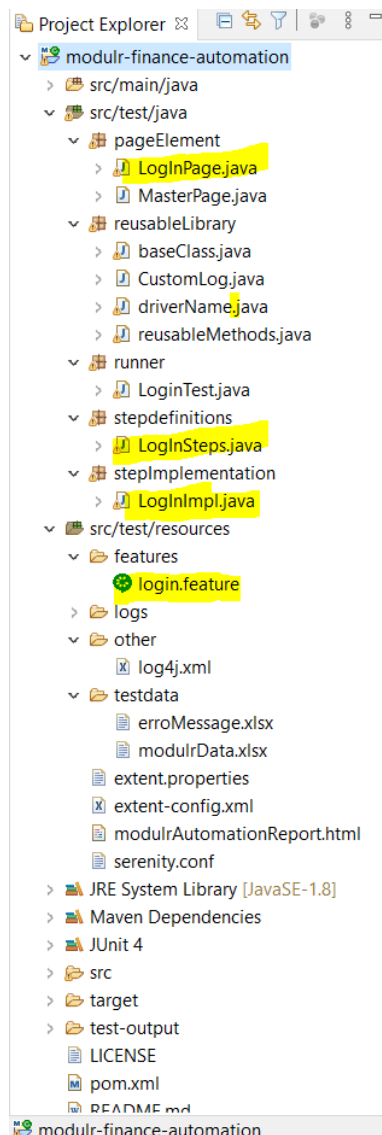
Demo framework creation and implementation is in progress and can be demoed on request.

*****GUI automation only. Can be created for API automation with difference in use of REST Assured Java library and validation again response than page objects.*

Home Assignments

1. Feature File (Login)

- Framework used is java + selenium with cucumber (Gherkin) for feature file
- Maven is used as dependency management and build tool.



- Login scenarios are written using below combination of users
 - o Sandbox user

- Developer user
- Direct Customers
- Partners
- Delegates
- Login Scenarios are
 - Valid Credentials
 - Incorrect UserName
 - Incorrect Password
 - Blank UserName & Password
 - Blank UserName
 - Blank Password
 - SignIn button Disable(Double Login)
 - Reset Password
 - Warning message to lock account for multiple incorrect passwords
- These test cases can be executed on different environment at same time by just changing value from Properties file (Customizable in framework).
 - Sandbox (URL : <https://secure-sandbox.modulrfinance.com/>)
 - dev1 (URL : <https://secure-dev.modulrfinance.com/>)
 - test1 (URL : <https://secure-test.modulrfinance.com/>)
- Tags are added in runner file to execute number of test cases based on purpose of testing.
 - MF-01 tag is used to test login feature for all combination of users during story testing.
 - Smoke/Sanity : This tag will be used only to test major functionalities due to small change in code
 - Regression: This tag will be used to include scenarios which are must for regression testing of application.

Detailed code is available in separate file attached with solution [login.feature]

Sample Screens:

login.feature 33 LoginSteps.java LoginImpl.java LoginTest.java modulr-finance-automation/pom.xml MasterPage.java baseClass.java LoginPage.java reusableMethods.java						
1	Feature: Modulr Login					
2						
3	# Environment to pass from Global Properties file and will be handled in logic. For timebeing its being send from feature file.					
4	@login @MF-001					
5	Scenario Outline: Modulr <customerType> logins with <scenario>					
6	Given "customerType" is valid Modulr user					
7	When user logins with credential from <test_data>					
8	scenario					
9	<scenario>					
10	Then login is "<loginStatus>" with "<outcome>"					
11						
12	@smoke @sanity					
13	Examples:					
14	TC_ID	customerType	scenario	result	outcome	test_data
15	TC_001	SandBox	Valid_Credentials	successful	loading account page	"data_001"
16	TC_002	SandBox	Incorrect_UserName	failed	error message	"data_001"
17	TC_003	SandBox	Incorrect_Password	failed	error message	"data_001"
18	TC_004	SandBox	Blank_UserName_And_Password	failed	error message	"data_001"
19	TC_005	SandBox	Blank_UserName	failed	error message	"data_001"
20	TC_006	SandBox	Blank_Password	failed	error message	"data_001"
21	TC_007	SandBox	Double_Login	failed	error message	"data_001"
22						
23	@regression					
24	Examples:					
25	TC_ID	customerType	scenario	result	outcome	test_data
26	TC_008	Developer	Valid_Credentials	successful	loading account page	"data_002"
27	TC_009	Developer	Incorrect_UserName	failed	error message	"data_002"
28	TC_010	Developer	Incorrect_Password	failed	error message	"data_002"
29	TC_011	Developer	Blank_UserName_And_Password	failed	error message	"data_002"
30	TC_012	Developer	Blank_UserName	failed	error message	"data_002"
31	TC_013	Developer	Blank_Password	failed	error message	"data_002"
32	TC_014	Developer	Double_Login	failed	error message	"data_002"
33	TC_015	Direct Customers	Valid_Credentials	successful	loading account page	"data_003"
34	TC_016	Direct Customers	Incorrect_UserName	failed	error message	"data_003"
35	TC_017	Direct Customers	Incorrect_Password	failed	error message	"data_003"
36	TC_018	Direct Customers	Blank_UserName_And_Password	failed	error message	"data_003"
37	TC_019	Direct Customers	Blank_UserName	failed	error message	"data_003"
38	TC_020	Direct Customers	Blank_Password	failed	error message	"data_003"
39	TC_021	Direct Customers	Double_Login	failed	error message	"data_003"
40	TC_022	Partners	Valid_Credentials	successful	loading account page	"data_004"
41	TC_023	Partners	Incorrect_UserName	failed	error message	"data_004"
42	TC_024	Partners	Incorrect_Password	failed	error message	"data_004"
43	TC_025	Partners	Blank_UserName_And_Password	failed	error message	"data_004"

```

54 login.feature
55 @login @MF-001
56 Scenario Outline: Modulr <customerType> recieves warning_for_account_to_get_locked_after_<noOfAttempts>_wornng_password_attempts
57 Given "customerType" is valid Modulr user
58 When user logins with credential from <test_data>
59 | scenario | noOfAttempts |
60 | <scenario> | <noOfAttempts> |
61 Then login is "<loginStatus>" with "<outcome>"
62
63 @smoke @sanity
64 Examples:
65 | TC_ID | customerType | noOfAttempts | scenario | loginStatus | outcome | test_data |
66 | TC_036 | SandBox | 5 | Wrong_Attempts | Paused | warningMessage | "data_001" |
67
68 @regression
69 Examples:
70 | TC_ID | customerType | noOfAttempts | scenario | loginStatus | outcome | test_data |
71 | TC_037 | Developer | 5 | Wrong_Attempts | Paused | warningMessage | "data_002" |
72 | TC_038 | Direct Customers | 5 | Wrong_Attempts | Paused | warningMessage | "data_003" |
73 | TC_039 | Partners | 5 | Wrong_Attempts | Paused | warningMessage | "data_004" |
74 | TC_040 | Delegates | 5 | Wrong_Attempts | Paused | warningMessage | "data_005" |
75
76 @login @MF-001
77 Scenario Outline: Modulr <customerType> should able to reset password
78 Given "customerType" is valid Modulr user
79 When user want to reset password
80 Then user should have provision to reset password
81 When user opt for reset pasword
82 Then reset password email should get trigger to users registered emailID
83
84 @smoke @sanity
85 Examples:
86 | TC_ID | customerType | scenario | result | test_data |
87 | TC_041 | SandBox | Reset_Password | Paused | "data_001" |
88
89 @regression
90 Examples:
91 | TC_ID | customerType | scenario | result | test_data |
92 | TC_042 | Developer | Reset_Password | Paused | "data_002" |
93 | TC_043 | Direct Customers | Reset_Password | Paused | "data_003" |
94 | TC_044 | Partners | Reset_Password | Paused | "data_004" |
95 | TC_045 | Delegates | Reset_Password | Paused | "data_005" |
96

```

2. Step Definition/Implementation (Login)

- Step Definition

Feature file will be glue to step definition where we will have the implementation of user behaviour. These will have detailed implementation under respective implementation file.

Detailed code for login is available in separate file attached with solution [LoginSteps.java]

Sample Screenshot from Editor:

```

3* import java.util.Map;
11
12 public class LoginSteps extends LogInImpl {
13
14     @Given("{string} is valid Modulr user")
15     public void is_valid_Modulr_user(String customerType) {
16         modulrData.put("customerType", customerType);
17     }
18
19     @When("user logins with credential from {string}")
20     public void user_logins_with_credential_from(String testCaseID, DataTable dataTable) {
21         modulrData.put("moduleName", "Login");
22         modulrData = loginWithUser(modulrData, testCaseID, dataTable);
23     }
24
25     @Then("login is {string} with {string}")
26     public void login_is_with(String result, String outcome) {
27         validateScenario(modulrData, result, outcome);
28         // Logout will be happen only if login is successful [Logic @PageLevel]
29         logout();
30     }
31
32     @When("user want to reset password")
33     public void user_want_to_reset_password() {
34         getResetPassword();
35     }
36
37     @Then("user should have provision to reset password")
38     public void user_should_have_provision_to_reset_password() {
39         validateResetPassword();
40     }
41
42     @When("user opt for reset pasword")
43     public void user_opt_for_reset_pasword() {
44         resetPassword();
45     }
46
47
48     @Then("reset password email should get trigger to users registered emailID")
49     public void reset_password_email_should_get_trigger_to_users_registered_emailID() {
50         validateDBForResetPasswordEmail();
51     }
52

```

- Step Implementation:

Step Implementation consist of detailed logic with all validation check and database validations.

Detailed code for login is available in separate file attached with solution [LoginImpl.java]

Sample Screenshot from Editor:

```
login.feature  LoginSteps.java  LoginImpl.java  LoginTest.java  modlr-finance-automation/pom.xml  MasterPage.java  baseClass.java  LoginPage.java  reusableMethods.java

22 public Map<String, String> loginWithUser(Map<String, String> modlrData, String testCaseID, DataTable dataTable) {
23     boolean shouldSignInClick = true;
24     modlrData = getDataTableValuesIntoHashMap(modlrData, dataTable);
25     testCaseID = modlrData.get("testCaseID");
26     TestData = getTestData(TESTDATA, getModuleSheetName(modlrData.get("moduleName")), testCaseID);
27     modlrData.put("userName", TestData.get("userName"));
28     modlrData.put("password", TestData.get("password"));
29     openUrl();
30     String scenario = modlrData.get("scenario").toLowerCase();
31
32     switch (scenario) {
33     case "valid_credentials":
34     case "incorrect_username":
35     case "incorrect_password":
36         enterUserName(modlrData.get("userName"));
37         enterPassword(modlrData.get("password"));
38     case "double_login":
39         clickSignIn();
40         clickSignInWithoutIsClickableChkAndValidate();
41         shouldSignInClick = false;
42         break;
43     case "blank_username_and_password":
44         break;
45     case "blank_username":
46         enterPassword(modlrData.get("password"));
47         break;
48     case "blank_password":
49         enterUserName(modlrData.get("userName"));
50         break;
51     case "reset_password":
52         shouldSignInClick = false;
53         clickResetPassword();
54         break;
55     case "wrong_attempts":
56         attemptWrongCredentialsLogin(modlrData);
57         shouldSignInClick = false;
58         break;
59     default:
60         enterUserName(modlrData.get("userName"));
61         enterPassword(modlrData.get("password"));
62     }
63     if (shouldSignInClick)
64         clickSignIn();
```

- Page Objects

This folder will have repository of all the page elements along with actions on these elements. We have used methods provided in serenity framework which is on top of selenium, to have additional in-built methods and features.

Detailed code for login is available in separate file attached with solution [LoginPage.java]

```
1 package pageElement;
2 import org.openqa.selenium.support.FindBy;
3 import net.serenitybdd.core.pages.WebElementFacade;
4 import reusableLibrary.CustomLog.LogInfo;
5
6 public class LoginPage extends MasterPage {
7
8     // ***** SignIn flow Starts Here *****//
9     @FindBy(xpath = "//input[@id='username-inp']")
10     WebElementFacade userNameInput;
11
12     public void enterUserName(String username) {
13         enterText(userNameInput, username);
14     }
15
16     @FindBy(xpath = "//input[@id='password-inp']")
17     WebElementFacade passwordInput;
18
19     public void enterPassword(String password) {
20         enterText(passwordInput, password);
21     }
22
23     @FindBy(xpath = "//button[@id='signInSubmitButton']")
24     WebElementFacade signIn;
25
26     public void clickSignIn() {
27         clickOnElement(signIn);
28     }
29
30     protected void chkSignInClickable() {
31         if (signIn.isClickable())
32             LogInfo.WithoutScreenShot("Sign In is multi-clickable and can send multiple login requests ", "Fail");
33         else
34             LogInfo.WithoutScreenShot("Sign In is not be multi-clickable to avoid multiple login requests ", "Pass");
35     }
36     // ***** SignIn flow Ends Here *****//
37
38     // ***** Forgot Password flow Starts Here *****//
39     @FindBy(xpath = "//*[@id='forgotPasswordHref']")
40     WebElementFacade forgotPassword;
41
42     public void clickResetPassword() {
43         clickOnElement(forgotPassword);
44     }
```

- Reusable Library

Reusable library will have methods that are used across framework and common everywhere.

E.g.

- Before performing any action on element check if it's available on screen.
- Logger to add custom messages on automation reports.

Flow will be like

Runner → Feature File → Step Definition → Step Implementation → Page Object →
Reusable Library → Base Class

- Customized Solution

This solution implemented has feature to rerun automation for any test environment just changing some config parameters from excel

BrowserForCurrentTest	chrome	
WebDriver_Name	WebDriver_Property	WebDriver_Path
chrome	webdriver.chrome.driver	C:\BhagyashriStudy\driver\chromedriver.exe
firefox	webdriver.gecko.driver	C:\BhagyashriStudy\driver\geckodriver.exe
IE	webdriver.ie.driver	D:\WebDriver\IEDriverServer.exe
Execution_Environment	sandbox	
sandbox	https://secure-sandbox.modulrfinance.com/	
From_MailID		
From_MailDPWD		
To_MailID		
dev_1	DevShakedown1	
test_2	TestShakedown1	
test_3	TestShakedown2	
env_3	TestShakedown3	

3. Login API Calls and analysis

Login API

Using sandbox account I am able to login into Modulr web and able to locate endpoints from Network tab of browser that got triggered one after the other.

Sr Num	API	URI
1	Authenticate	https://api-sandbox.modulrfinance.com/api-sandbox/security-service/authenticate
2	MFAStatus	https://api-sandbox.modulrfinance.com/api-sandbox/security-service/mfastatus/e71531a-84b44e853446-03e61a-99a72f0
3	Customers	https://api-sandbox.modulrfinance.com/api-sandbox/customers?
4	Direct Debit	https://api-sandbox.modulrfinance.com/api-sandbox/accounts?identifierType=DD
5	Accounts	https://api-sandbox.modulrfinance.com/api-sandbox/accounts?page=0&size=15&sortField=id&sortOrder=asc&q=
6	Revoke	https://api-sandbox.modulrfinance.com/api-sandbox/security-service/revoke
7	Audit	https://api-sandbox.modulrfinance.com/api-sandbox/audits

Detailed analysis with below data point for authentication are added in attached excel with this document—

- URI
- Request Header
- Body Payload
- Response
- Logic of sharing token and algorithm used as public & private encryption.

API Details is available in separate file attached with solution [\[API_And_ScenarioSheet.xlsx\]](#)

Additional API's

Along with sign In API, I went through below additional endpoints to get brief idea about Modlur API (URI/Header/Payload)

APIs	EndPoints
AccessGroup	4
account-name-check	1
Accounts	8
Address Lookup	1
Async	1
Beneficiaries	3
Cards	14
Customers	4
Direct Debit	
Outbound	3
Direct Debits	8
File Upload	3
Inbound payments	1
Notifications	5
Payments	4
PISP Service	3
Rules	5
Transactions	1
Authentication	4
Grand Total	73

Supporting Skills

- Skills matching with Job description
 - Manual/Functional Testing
 - Writing test scenarios/test cases and executing them on different browsers/mobile devices.
 - Reporting findings in test management tool(Jira) and track till closure
 - Automation
 - Develop/Customize framework for API/GUI automation
 - Create/update/execute automation scripts in Java / Java Script using developed automation framework.
 - Git/GitHub as version control
 - Basic knowledge on Jenkins
 - Supporting skills
 - Good in Problem solving, proactive
 - Can work under pressure in strict timelines
 - Good team player
 - Worked in agile projects
 - Good at creating documentations

Framework Developed

1. CodeElan/STA(Eleas) BDD framework (Java + Selenium +Cucumber+Maven+Reporting)
[API + GUI]
2. Demo JS BDD framework (Protractor + HTML reporting)
3. Test Data driven framework using SoapUI (CountryInfo Service)
4. Hybrid framework using SoapUI (CountryInfo Service)
5. Used SpringBoot framework to customise Student API (Get/POST/PUT/Delete) for AT

It would be great opportunity for me if requested for demo. Happy to arrange demo and detailed walkthrough on this automation activity.

My Skillsets

Editors & Programming Languages



Frameworks



Testing Tools



References/ Recommendations:

Based on request happy to share references.